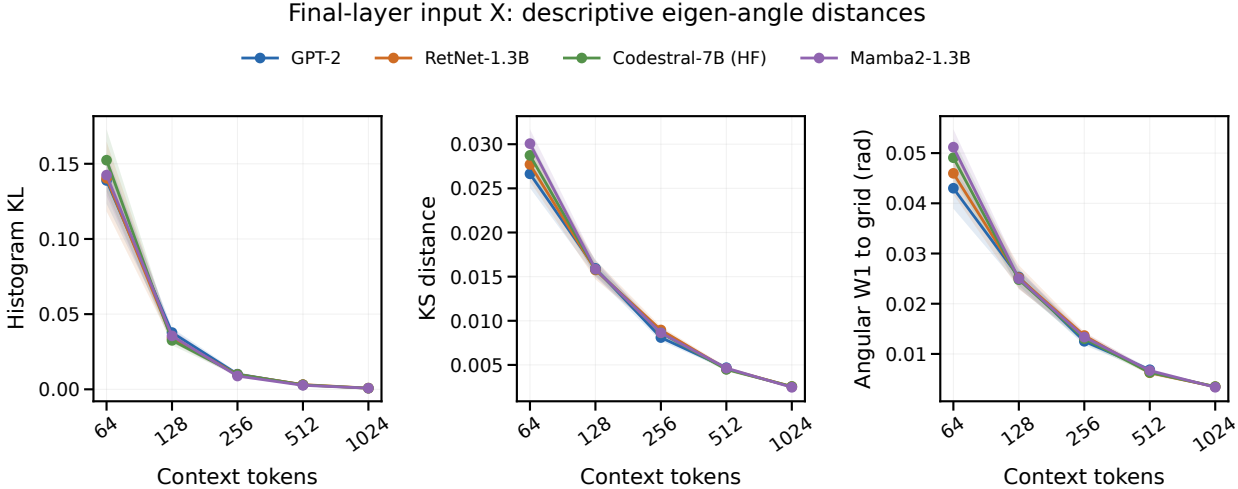
Final-layer input X: descriptive eigen-angle distances
GPT-2
RetNet-1.3B
Codestral-7B (HF)
Mamba2-1.3B
Histogram KL
0.15
0.10
0.05
0.00
KS distance
0.030
0.025
0.020
0.015
0.010
0.005
Angular W1 to grid (rad)
0.05
0.04
0.03
0.02
0.01
64
128
256
512
1024
Context tokens
32 documents; numerically full-row-rank X only. Shading: 95% document-bootstrap intervals.